

Abstract

An analysis was conducted to identify student profiles based on personality traits and discourse characteristics that could predict the likelihood of persuading a debate partner. Results suggested that students who successfully persuaded their partners tended to be more extroverted and displayed desirable discourse characteristics, particularly politeness. These students also appeared able to inhibit a potential tendency toward rudeness, which may be theoretically related to the higher impulsivity often associated with extroversion.

Objective

To analyze the relationship between students' personality characteristics, discourse features, and their ability to persuade another student during an academic debate.

Variable Definitions

Independent (Predictor) Variables

Extraversion.

Extraversion refers to an individual's level of social engagement and sociability. It is generally considered to have a substantial genetic component. Individuals with high levels of extraversion are typically motivated to seek social interaction and often perform better in group activities. They also tend to be more impulsive. In contrast, individuals with low levels of extraversion (introverts) generally prefer solitary activities or interactions with a small number of people. Unlike extraverts, group activities may be psychologically draining for them, leading them to favor more reflective and intellectually oriented activities.

Number of Arguments.

The total number of arguments produced by a student during the interaction. This variable was examined because argumentative behavior is often assumed to influence persuasive success in academic contexts.

Number of Connectors.

Connectors are words and phrases such as "because" or "in this sense." They were examined because they are considered indicators of discourse quality in both spoken and written communication. Greater discourse clarity may contribute to persuasive effectiveness.

Politeness Indicators.

These include words and expressions such as "thank you," "you're welcome," and discourse softeners such as "I think" or "it seems to me." Such expressions reduce the perception that a speaker is imposing a viewpoint on another person and are generally considered beneficial in persuasive interactions.

Rudeness Indicators.

In contrast to politeness indicators, these are expressions such as "we have to" or "we should," which may create the impression that the listener has little choice but to accept the speaker's position.

Powerless Language Indicators.

These indicators reflect hesitation or uncertainty in speech. Examples include fillers and vocalizations such as "uh..." or "mhm...". This variable has received comparatively little research attention, and there is currently no consensus regarding its impact on persuasive outcomes.

Dependent Variable (targett)

Persuasion Success.

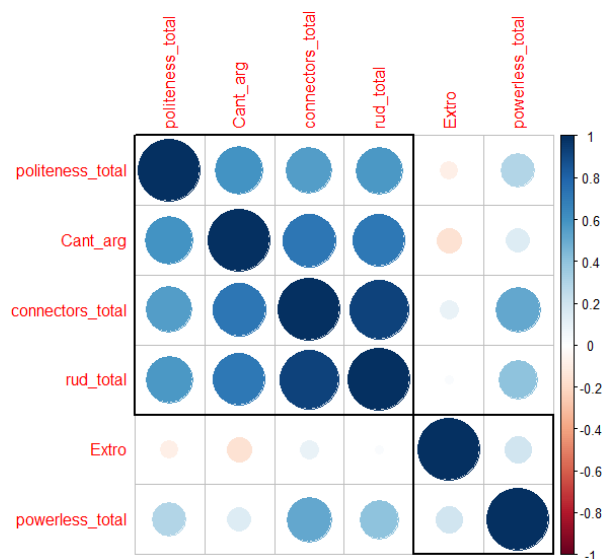
Whether a student successfully convinced their debate partner that their position was the correct one.

Sample and Experimental Procedure

The sample consisted of 49 psychology students from two universities located in Rosario, Argentina, with a mean age of 21 years. Participants were asked to debate with a partner about the use of ChatGPT for university academic tasks. Each student was randomly assigned a position ("in favor" or "against") and was instructed to persuade their partner that their assigned position was the correct one.

Exploratory Analysis

Figure 1. Variable correlation matrix

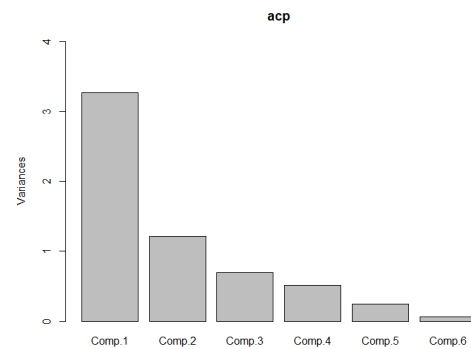


As shown in Figure 1, extraversion displayed weak negative relationships with the number of arguments ($r = -0.16$) and politeness indicators ($r = -0.08$), and a weak positive relationship with powerless language indicators ($r = 0.19$). In contrast, the discourse-related variables were strongly associated with one another, as expected.

Table 1. Principal Components

Metric	Comp.1	Comp.2	Comp.3	Comp.4	Comp.5	Comp.6
Standard deviation	1.809004	1.1029056	0.8323819	0.7186178	0.49086977	0.24673659
Proportion of Variance	0.545416	0.2027335	0.1154766	0.0860686	0.04015885	0.01014649
Cumulative proportion	0.545416	0.7481495	0.8636261	0.9496947	0.98985351	1

Figure 2. Principal Components



Following the minimum variance criterion ($SD > .75$), Components 1, 2, and 3 were retained for further examination. However, Component 4 was also briefly inspected because it appeared relatively distinct from Component 5 and presented a potentially meaningful theoretical profile.

Figure 3. Correlations Between the Original Variables and the Principal Components



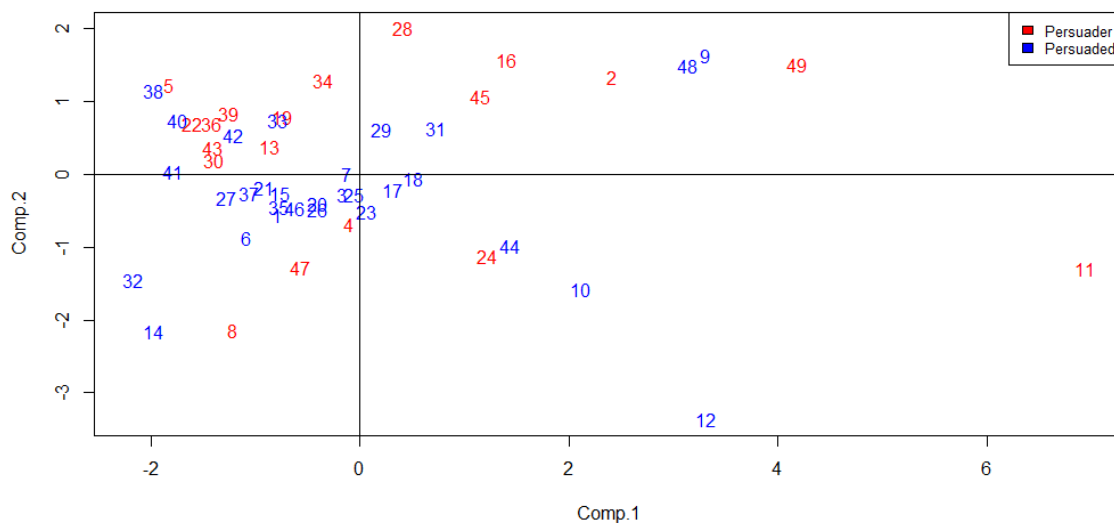
The relationship between the original variables and the principal components suggested that the first component was primarily composed of discourse-related variables. This component may represent students who displayed desirable discourse characteristics, although it also included rudeness indicators. Notably, it appeared largely independent of extraversion.

The second component was characterized by high levels of extraversion. Students with high scores on this component tended to produce fewer arguments and more powerless language indicators. In addition, a small positive association with rudeness indicators was observed. Taken together, these patterns may suggest that highly extroverted students displayed comparatively weaker discourse skills.

The third component appeared to represent what could be described as "moderate" students. These individuals showed moderate levels of extraversion, produced some arguments, and displayed a strong tendency to avoid powerless language, as reflected by the substantial negative correlation with this variable. This profile may indicate students who combined moderate extraversion with relatively strong discourse abilities.

Finally, the fourth component appeared to represent students who prioritized politeness during the interaction. This interpretation is supported by the strong positive association with politeness indicators and the negative association with rudeness indicators. The weak relationship with extraversion may suggest that these students tended to be more introverted. Furthermore, the absence of a relationship with the number of arguments may indicate that their primary goal was not necessarily persuasion, but rather maintaining a cordial and respectful interaction. Although Component 4 did not satisfy the variance-based selection criterion, it was retained for brief examination because it represented a theoretically meaningful student profile.

Figure 4. Participant Distribution Across the First Two Principal Components According to Persuasion Outcome

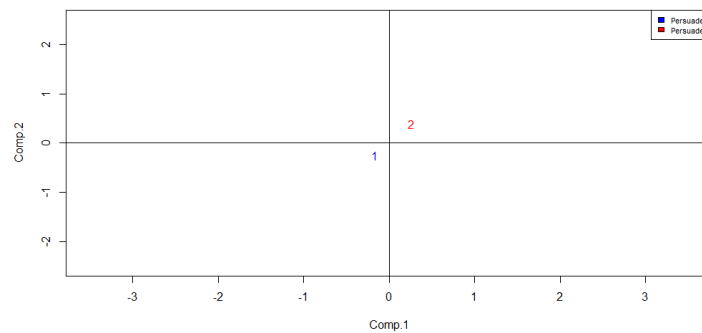


The comparison between Components 1 and 2 revealed several differences between students who successfully persuaded their partner (persuader) and those who did not (persuaded). Most of the students who succeeded in persuading their partner were located in the upper quadrants of the plot, particularly in the positive region of Component 2, which was interpreted as representing higher levels of extraversion. In addition, many of these students also scored positively on Component 1, suggesting the presence of relatively strong discourse skills.

In contrast, students who failed to persuade their partner were predominantly located in the negative region of Component 2, indicating lower levels of extraversion. Many of these students also obtained lower scores on Component 1, suggesting weaker discourse abilities.

These findings may indicate that extroverted students with strong discourse skills are more likely to persuade others. However, the results also suggest that discourse skills alone may not be sufficient. A certain degree of extraversion appears to be an important characteristic, as several students with relatively modest discourse skills but high extraversion scores were still able to persuade their partner. In this sense, extraversion may partially compensate for weaker discourse abilities in persuasive interactions.

Figure 5. Principal Component Analysis Centroids According to Persuasion Outcome



The centroid analysis is consistent with the proposed hypothesis. In addition, the comparison between students who successfully persuaded their partner and those who did not revealed a statistically significant difference on Component 2.

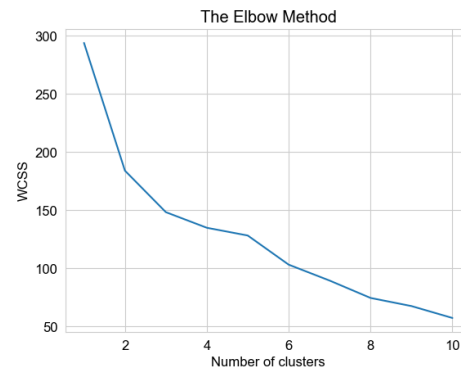
Table 2. Comparison of Principal Component Scores According to Persuasion Outcome

Components	T-value
1	-0.72
2	-1.98*
3	0.26
4	0.93

* $p < .06$

To obtain a more detailed understanding of the data, a k-means cluster analysis was conducted using the independent variables (extraversion, arguments, connectors, politeness, rudeness, and powerless language indicators). The scores of Components 1 and 2 were then used as the axes for visualizing the resulting clusters within the principal component space.

Figure 6. Elbow Plot for K-Means Clustering



The elbow method suggested a solution of either four or five clusters. However, because the additional cluster contained only a small number of cases, the four-cluster solution was retained for subsequent analyses.

Figure 7. Scatter Plot of Principal Components and Resulting Clusters

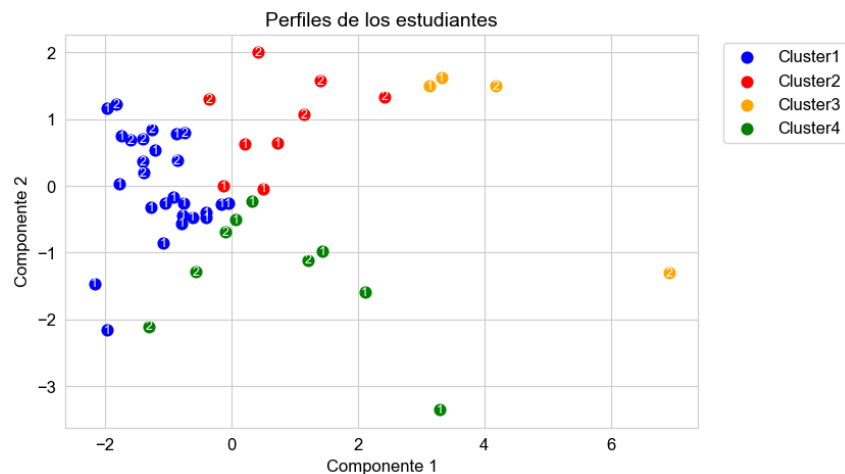


Figure 7 highlights Cluster 2, which is associated with positive values on Component 2 and, simultaneously, with positive values on Component 1. This pattern suggests a combination of extraversion and desirable discourse characteristics, which may contribute to a greater ability to persuade others.

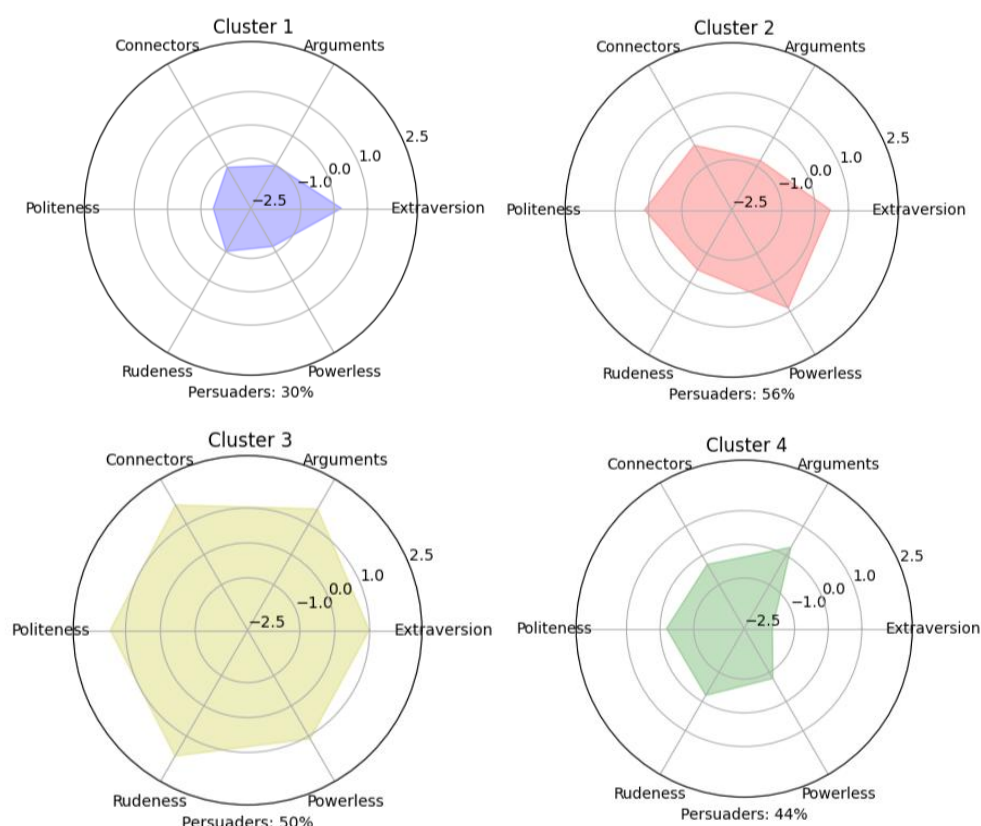
To facilitate interpretation, the mean score of each cluster was calculated for every variable, along with the number of students who successfully persuaded their partner and those who did not. In addition, radar plots were created for each cluster to provide a visual summary of their profiles.

Table 3. Composition of Each Cluster Across Variables

	Cluster 1		Cluster 2		Cluster 3		Cluster 4	
Variables	M	%	M	%	M	%	M	%
Extraversion	31.26	-	32.67	-	35.75	-	20.33	-
Arguments	4.74	-	6.00	-	19.25	-	12.22	-

Connectors	20.07	-	52.11	-	126.5	-	50.78	-
Politeness	3.70	-	8.00	-	11.75	-	7.00	-
Rudeness	10.67	-	22.56	-	65.5	-	27.22	-
Powerless								
Language	2.67	-	9.11	-	9.75	-	3.89	-
Persuaders	-	30%(8)	-	56%(5)	-	50%(2)	-	44%(4)
Persuaded	-	70%(19)	-	44%(4)	-	50%(2)	-	56%(5)

Figure 8. Radar plots for each cluster across all variables.



Note. Radar plot scores are based on the standardized cluster means for each variable.

The percentages of participants who successfully persuaded their partners (Table 3) should be interpreted cautiously due to the small number of cases within some clusters. Cluster 2 showed the highest proportion of successful persuaders. This cluster was associated with positive values on Component 2, which appeared to represent extraversion, and positive values on Component 1, which represented discourse-related indicators. These findings are consistent with the previous interpretation that extraverted students with strong discourse skills may be better equipped to persuade their partners.

In addition, the radar plot (Figure 8) suggests that politeness was the discourse feature that most clearly distinguished Cluster 2. This may indicate that politeness was particularly important for extraverted students' persuasive success. Interestingly, Cluster 1 appeared slightly more

extraverted than Cluster 2 but displayed lower levels of politeness, suggesting that extraversion may partially compensate for weaker discourse skills.

Cluster 3 (Figure 8) showed the highest average level of extraversion but also exhibited high levels of rudeness indicators. This pattern may further support the idea that politeness is a key factor that enhances the persuasive effectiveness of extraverted individuals. However, because Cluster 3 contained very few cases, these findings should be interpreted with caution.

Cluster 4 appeared to be the least successful cluster in terms of persuasion. The contrast between Cluster 2, hypothetically the most effective profile, and Cluster 4, hypothetically the least effective, may be particularly informative. As shown in Table 4, Cluster 2 displayed higher levels of extraversion, higher levels of politeness, and lower levels of rudeness than Cluster 4.

Table 4. Mean differences between Cluster 2 and Cluster 4 in Extraversion, Politeness, and Rudeness

	Cluster 2		Cluster 4		Cluster 2 vs 4
	M	SD	M	SD	T-student
Extraversion	32.67	5.81	20.33	4.27	5.13*
Politeness	8.00	2.50	7.00	7.07	0.41
Rudeness	22.56	10.11	27.22	11.16	-0.92

* $p < .01$

(Analysis not included in the code.)

As suggested by previous findings, indicators of uncertainty (Powerless language) do not appear to substantially affect persuasion outcomes, although additional research is needed before drawing firm conclusions (analysis not included in the code).

Subsequently, Component 2, whose interpretation had already been explored, was used to build a logistic regression model in order to evaluate whether persuasion outcomes could be predicted from the characteristics represented by this component. The next section presents the results of this analysis.

Table 5. Logistic Regression Model: Component 2 as a Predictor of Persuasion

Métrica	Valor
β_1	0.60
Std.Error	31
Odds Ratio	1.82
Z-value	1.91*
χ^2	10.87**
AIC	65.18

* $p < .06$, ** $p > .05$

The logistic regression model was statistically significant (Table 5). For each one-unit increase in Component 2, the odds of persuading a partner increased by a factor of 1.82. This corresponds to an estimated probability of approximately 64% that a student with these characteristics would

successfully persuade their partner. Furthermore, the Hosmer–Lemeshow goodness-of-fit test ($p > .05$) indicated no evidence of poor model fit.

As previously observed, positive values on Component 2 (Figure 4) were associated with higher levels of extraversion and were more frequently observed among students who successfully persuaded their partners. These students were primarily represented by Cluster 1 (Figure 7). Cluster 2 (Figure 7) shared the same characteristic but also exhibited stronger discourse-related skills. Therefore, Clusters 1 and 2 combined should theoretically provide a better prediction of persuasion outcomes than Component 2 alone (Figure 4). This effect might be even stronger when considering only students belonging to these clusters.

To explore this possibility, the two groups described above were used to build additional logistic regression models and compare their predictive performance with the initial model based solely on Component 2 (Table 5).

Table 6. Comparison of Logistic Regression Models: Clusters 1 and 2 vs. Clusters 1 and 2 with Positive Component 2 Scores Only

Model	$\beta 1$	DE	Odds Ratio	z-value	x2	AIC
Clusters 1 & 2	3.21	1.07	24.85	3.01*	3.04*	30.20
Clustes 1 & 2 Positive	2.09	1.25	8.10	1.67***	7.78**	28.20

* $p < .01$, ** $p > .05$, *** $p > .10$

The model based on Clusters 1 and 2 using Component 2 scores was statistically significant, whereas the model restricted to students from Clusters 1 and 2 with only positive Component 2 scores did not reach statistical significance (Table 6). Despite the lack of statistical significance, the lower Akaike Information Criterion (AIC) value observed in the second model suggests a potentially better balance between model fit and complexity, although the estimated odds ratio was substantially smaller.

These findings are generally consistent with the previous results. The second model (Clusters 1 and 2 with positive Component 2 scores only; Table 6) may not outperform the first model because extraversion alone does not appear to be sufficient for successful persuasion. Instead, extraversion seems to be more effective when combined with strong discourse-related skills.

Furthermore, the second model may suggest that very high levels of extraversion are not necessarily advantageous and could even be counterproductive for persuasion. This interpretation is theoretically plausible and is also supported by the present data, as higher levels of extraversion may be associated with greater use of rude discourse markers. Such markers contrast with indicators of politeness, which were previously identified as one of the discourse characteristics most strongly associated with successful persuasion when combined with extraversion. Finally, the classification performance of both models was evaluated:

Table 7. Classification Performance of the Clusters 1 and 2 Model versus the Clusters 1 and 2 Positive Component 2 Model

	Clusters 1 & 2	Clusters 1 & 2 Positive
Accuracy	.80	.57
Precision	.71	.62
Recall	.77	.77
F1-Score	.74	.69

Regarding classification performance, the model based on Clusters 1 and 2 outperformed the model restricted to positive Component 2 scores. The two models showed similar sensitivity, meaning that they were equally capable of identifying true positive cases. However, the Clusters 1 and 2 model achieved higher accuracy, precision, and F1-score, indicating superior overall predictive performance.

Conclusion

Based on the obtained results, considering both the estimated odds ratios (Table 6) and the fact that the Clusters 1 and 2 model achieved higher accuracy and F1-score than the model restricted to positive Component 2 scores (Table 7), it can be concluded that the first model provides a better representation of the student profile associated with successful persuasion.

This profile is characterized by students who are relatively extroverted and possess strong discourse-related skills. Among these skills, politeness appears to play a particularly important role, while the inhibition of rude discourse may also be relevant. This tendency may be linked to the greater impulsivity often associated with extraversion. When such impulsive tendencies are controlled and accompanied by polite communication, the resulting profile appears to have the highest probability of persuading others.

Limitations

The data were obtained from an experimental design that was originally part of a broader research activity. In addition, several variables were constructed specifically for the purposes of this project, with the exception of extraversion, argument production, and persuasion outcome. Nevertheless, these variables may warrant further investigation in future studies.

Another limitation concerns the potential dependence among observations. Participants interacted with one another during the debates, meaning that observations may not be entirely independent. However, this dependence is unlikely to artificially strengthen the present findings. If anything, it may work against the proposed hypothesis. The characteristics examined in this study—particularly extraversion—are generally assumed to be relatively stable over time. Although some discourse-related characteristics, such as politeness, may change through learning and experience, substantial changes are unlikely to occur during the approximately nine-minute debates analyzed here.

Furthermore, interaction-based dependence may introduce noise into the analysis. For example, if two highly extroverted and polite students happened to be randomly paired together, one would still be classified as unsuccessful in persuasion simply because only one participant could persuade the other. In such cases, observations with very similar characteristics would receive different

outcome labels, potentially obscuring the relationship between the predictors and persuasion success.

The discourse variables were constructed using a bag-of-words approach based on findings from previous research and methods previously employed for the automatic detection of discourse indicators. Among these variables, politeness, rudeness, and connectors have received the greatest empirical attention, although connectors have rarely been examined within persuasive contexts. Their relationship with persuasion is therefore primarily theoretical. While discourse clarity has been associated with persuasion in previous research, connectors themselves have not been directly studied as predictors of persuasion in spoken language.

Powerless language has also been investigated in prior studies; however, as discussed earlier, there is still no clear consensus regarding its effect on persuasion outcomes. Finally, the argument variable was constructed through the automatic classification of conversational turns using a bag-of-words approach based on findings from an exploratory study conducted by the present author on argumentative discourse markers.

The present analysis did not explore other discourse variables in depth because politeness appeared to be the most relevant discourse feature associated with successful persuasion, while rudeness appeared to be a characteristic worth avoiding. Based on the results of this study, particularly the initial component analysis (Figure 4), it may be hypothesized that a sufficient number of arguments is necessary for persuasion. However, argument quantity alone did not appear to be the factor that distinguished students who successfully persuaded their partners from those who did not.